Geonwoo Kim

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Education

University of Chicago 2025–Present
Graduate Student at Large, Department of Economics
PhD coursework: ECON 31000 Empirical Analysis I (in progress).

Grinnell College 2019–2024
B.A. in Mathematics with Honors; Concentration in Statistics.
Selected coursework: Real Analysis, Numerical Analysis, Mathematical Statistics, Advanced Econometrics, Measure Theory, Measure-Theoretic Probability, Empirical Process Theory.

Fields

Econometric theory High-dimensional inference, measurement error, computational econometrics.
Applied econometrics Policy evaluation, labor economics, economics of AI.

Research

1. *Denoised Partial Correlation for Conditional Independence Tests under Noisy Confounders* (with Suyong Song and Jonathan Wells). Under revision.
2. *Inference for Treatment Effects with Mismeasured High-Dimensional Controls* (with Suyong Song). Presented at Midwest Econometrics Group 2024; under revision.
3. *Signal Space Conjugate Gradient Iterative Hard Thresholding* (with Kate Bartz and Jeffrey Blanchard). Under review.
4. *Inference After AI Measurement* in progress.

Experience

Research Professional in Econometric Theory 2025.07–Present
University of Chicago, Department of Economics

Inference for Linear Systems with Unknown Coefficients
PIs: Yuehao Bai, Kirill Ponomarev, Andres Santos, Azeem Shaikh, Max Tabord-Meehan, Alexander Torgovitsky
- Work on tests for whether a data-generating process lies in sets defined by large systems of linear restrictions, with applications to nonparametric IV and shape restrictions.
- Contribute to uniform Gaussian and bootstrap approximations for supremum-type statistics; implement procedures in R, solve linear and mixed-integer programs, and run parallelized Monte Carlo experiments.

The Identifying Power of an Assumption: Comparing Sharp Identified Sets under Linear Restrictions
PIs: Yuehao Bai, Shunzhuang Huang, Azeem Shaikh, Edward Vytlacil
- Audit and extend Python code for a computational partial-identification project comparing sharp identified sets under alternative linear restrictions in finite-dimensional models.
- Replace sampled or heuristic dual-piece discovery with exact polyhedral enumeration using cdd/pycddlib; verify endpoints, redundancy diagnostics, pruning routines, ordinal-IV computations, and replication-package output.

Inference for Policy Learning and Optimal Welfare
PIs: Kirill Ponomarev and Vira Semenova
- Assist with computational and theoretical components of a policy-learning project on welfare bounds and confidence bands for data-driven treatment rules.
- Implement simulation and replication code, check analytical derivations, and diagnose finite-sample performance of proposed inference procedures.

Undergraduate Researcher 2024.08–2024.12
Grinnell College, Department of Statistics
Advisor: Jonathan Wells
- Used random matrix theory and representation theory to develop an efficient statistic for a permutation-based conditional-independence test under noisy confounders.
- Formulated an empirical risk minimization procedure that minimizes rank mismatch in estimated residuals; implemented the algorithm in R and ran simulations.

Undergraduate Researcher 2023.08–Present
University of Iowa, Department of Economics
Advisor: Suyong Song

- Developed a high-dimensional errors-in-variables regression procedure based on an orthogonal score that mitigates measurement error and regularization bias in nuisance estimation.
- Constructed a measurement-error covariance estimator using auxiliary information and the empirical spectral distribution; derived the influence function for asymptotic variance and uniform inference.
- Ran large-scale Monte Carlo simulations showing improved coverage and reduced omitted-variable bias relative to naive estimators.

Undergraduate Researcher 2023.05–Present
Grinnell College, Department of Mathematics
Advisor: Jeffrey Blanchard

- Developed Signal Space Conjugate Gradient Iterative Hard Thresholding, a sparse optimization algorithm for dictionary-domain signal recovery.
- Proved convergence and noise-stability results under bounded noise and alternative dictionary choices; implemented and benchmarked SSCGIHT in Python.

Research Assistant 2022.03–2024.03
Seoul National University, Department of Operations Research
PI: Song-Hee Kim

- Studied the causal impact of a shift-choice system on nurse turnover and service quality using administrative shift rosters, turnover records, and quality indicators from a large hospital.
- Constructed a nurse-month panel; inferred department-level treatment timing under staggered adoption; assisted with generalized DiD, event-study, robustness, and sensitivity analyses.

Research Assistant 2019.02–2019.06
Science and Technology Policy Institute, South Korea
PI: Minhyung Lee

- Conducted policy research on cross-institutional public R&D collaboration networks; implemented graph-based algorithms to identify mediator institutions in innovation diffusion.

Teaching

Grinnell College

STA 336: Probability and Statistics I, Course Mentor — Fall 2024
ECN 280: Microeconomic Analysis, Course Grader — Fall 2024
STA 295: Statistical Learning, Course Mentor — Spring 2024
ECN 286: Econometrics, Course Mentor — Fall 2023, Spring 2024

Hosanna International School, Busan, South Korea

High School Calculus, Teaching Assistant — 2020.05–2020.08

Service

Korean Students Association President, Grinnell College — 2022–2023
Squad Leader, Republic of Korea Army — 2021–2022
Drill Sergeant, Republic of Korea Army — 2020–2022

Skills

Programming Python, R, MATLAB, Stata/Mata, LaTeX, Git.
Languages Korean (native), English (fluent), German (intermediate).